

Conference Paper Title*

*Note: Sub-titles are not captured by IEEE Xplore

Konstantina Koulaktsidou

Department of Electrical and Computer Engineering
National Technical University of Athens
email

Fragiska Kiminou

Department of Electrical and Computer Engineering
National Technical University of Athens
email

Nikolaos Moraitis

Department of Electrical and Computer Engineering
National Technical University of Athens
email

Abstract—Graph embedding techniques are important tools for representing graphs in low-dimensional vector spaces, enabling effective downstream tasks. In this work, we evaluate unsupervised and supervised representative whole graph embedding techniques on real world benchmark datasets. Specifically, we compare Graph2Vec, NetLSD, and the Graph Isomorphism Network (GIN) across three datasets (MUTAG, ENZYMES, and IMDB-MULTI). Performance is assessed through graph classification and clustering tasks, considering predictive accuracy, computational efficiency and scalability. Our experiments indicate that GIN achieves the highest accuracy but requires substantial computational resources. Graph2Vec balances performance and efficiency, while NetLSD provides competitive clustering performance with minimal overhead. Therefore it is apparent that each method has its distinct trade-offs. All methods maintain reasonable robustness under moderate perturbations, with sensitivity depending on dataset characteristics. These findings provide practical insights that aid in selecting graph embedding approaches based on the task requirements, dataset properties and computational constraints.

Index Terms—Graph2Vec, NetLSD, GIN, Graph Embedding, Graph Classification, Graph Clustering

I. INTRODUCTION

Graphs provide a powerful representation for complex relational data and are widely used in domains such as network analysis, bioinformatics and chemistry. In many scenarios, the learning task is defined at the graph level, such as classifying molecular graphs by biological activity. However, graphs have irregular structure with variable sizes, unordered nodes and complex connectivity patterns, features that make them face challenges to traditional machine learning algorithms, which typically require fixed dimensional vector inputs.

Graph embedding methods address this challenge by mapping entire graphs into continuous vector representations that preserve the structural and semantic infor-

mation of the graph. These kinds of representations enable the use of standard machine learning techniques for downstream tasks such as graph classification, clustering and similarity search. As a result, graph embeddings have successfully become a core component of modern graph based learning pipelines.

Early approaches relied on handcrafted features or graph kernels, which often suffer from high computational complexity or limited expressiveness [CITATIONS NA VROYME]. Recent approaches automatically learn graph representations and can be categorized into unsupervised[CITATIONS] and supervised techniques[CITATIONS]. Unsupervised methods learn graph representations without relying on labeled data exploiting often structural patterns. In this category belong Graph2Vec [CITATION], which adapts the skip-gram model to learn graph level embeddings from rooted substructures, and NetLSD, which uses heat kernel signatures derived from the graph Laplacian to capture global topological characteristics. On the other hand, supervised procedures, which are based on graph neural networks (GNNs) [CITAITON], learn task specific embeddings through end-to-end training. Among these, the Graph Isomorphism Network (GIN) [CITATION] has gained significant attention due to its strong theoretical expressiveness. GIN can learn highly discriminative graph representations while achieving superior performance on graph classification benchmarks. However, choosing this type of procedure typically comes with the cost of increased computational complexity, higher memory consumption and the need for labeled training data, which is often not provided to us.

Despite the development of graph embedding techniques[CITATIONS], comparisons between unsupervised and supervised methods remain limited[CITATIONS]. Many existing studies focus primarily on classification accuracy and overlooking significant considerations such as computational cost, embedding dimensionality and robustness to structural permutations. These are critical when working with real world small datasets (EMB1),

where overfitting, instability and minimum training data can easily affect the model’s performance.

In this work, we focus on EMB1 and we evaluate and compare unsupervised and supervised graph embedding methods on small benchmark datasets from the TUDataset collection: MUTAG, ENZYMES, and IMDB-MULTI. Studying small datasets is as meaningful as big ones, as they represent realistic scenarios in domains such as chemistry and biology, where labeled data is costly to obtain and sometimes no where to be found and model robustness is essential. From the unsupervised category, we use Graph2Vec and NetLSD and from the supervised GIN. To ensure fairness, all methods are evaluated using a unified experimental pipeline with minimal difference from GIN. The evaluation extends beyond standard classification and includes clustering behavior, computational efficiency and stability under random perturbations in the graphs structure.

The main contributions are summarized as follows:

- We present a comparative evaluation of unsupervised and supervised whole-graph embedding methods on widely used real-world datasets,
- We analyze the trade-off between classification accuracy and computational cost, highlighting practical considerations for method selection,
- We perform a stability analysis under structural permutations, assessing the robustness of different embedding paradigms

II. RELATED WORK

Briefly discuss prior graph embedding techniques (unsupervised vs. supervised).

Mention previous benchmarks and known trade-offs.

III. METHODS

Graph2Vec: how it captures subgraph patterns.

NetLSD: spectral descriptors, computational efficiency.

GIN: neighborhood aggregation, supervised learning approach.

Implementation details: libraries used (KarateClub, PyTorch Geometric), embedding dimensionality, training setup

IV. DATASETS

Describe MUTAG, ENZYMES, IMDB-MULTI: type of graphs, nodes, labels.

Any preprocessing steps.

V. EXPERIMENTS AND EVALUATION

A. Classification

SVM/MLP classifier for embeddings.

Report accuracy, F1-score, AUC.

Computational metrics: training time, generation time, memory usage.

Analysis of embedding dimensionality vs. performance.

B. Clustering

k-means/spectral clustering.

ARI scores and qualitative visualization (t-SNE or UMAP).

Discussion of cluster separation quality.

C. Stability Analysis

Random perturbations (edges/nodes).

Compare embedding changes and drop in classification accuracy.

VI. RESULTS

Tables and figures summarizing all tasks.

Highlight trade-offs: performance vs. efficiency vs. stability.

REFERENCES

Please number citations consecutively within brackets [1]. The sentence punctuation follows the bracket [2]. Refer simply to the reference number, as in [3]—do not use “Ref. [3]” or “reference [3]” except at the beginning of a sentence: “Reference [3] was the first ...”

Number footnotes separately in superscripts. Place the actual footnote at the bottom of the column in which it was cited. Do not put footnotes in the abstract or reference list. Use letters for table footnotes.

Unless there are six authors or more give all authors’ names; do not use “et al.”. Papers that have not been published, even if they have been submitted for publication, should be cited as “unpublished” [4]. Papers that have been accepted for publication should be cited as “in press” [5]. Capitalize only the first word in a paper title, except for proper nouns and element symbols.

For papers published in translation journals, please give the English citation first, followed by the original foreign-language citation [6].

REFERENCES

- [1] G. Eason, B. Noble, and I. N. Sneddon, “On certain integrals of Lipschitz-Hankel type involving products of Bessel functions,” *Phil. Trans. Roy. Soc. London*, vol. A247, pp. 529–551, April 1955.
- [2] J. Clerk Maxwell, *A Treatise on Electricity and Magnetism*, 3rd ed., vol. 2. Oxford: Clarendon, 1892, pp.68–73.
- [3] I. S. Jacobs and C. P. Bean, “Fine particles, thin films and exchange anisotropy,” in *Magnetism*, vol. III, G. T. Rado and H. Suhl, Eds. New York: Academic, 1963, pp. 271–350.
- [4] K. Elissa, “Title of paper if known,” unpublished.
- [5] R. Nicole, “Title of paper with only first word capitalized,” *J. Name Stand. Abbrev.*, in press.
- [6] Y. Yorozu, M. Hirano, K. Oka, and Y. Tagawa, “Electron spectroscopy studies on magneto-optical media and plastic substrate interface,” *IEEE Transl. J. Magn. Japan*, vol. 2, pp. 740–741, August 1987 [Digests 9th Annual Conf. Magnetism Japan, p. 301, 1982].
- [7] M. Young, *The Technical Writer’s Handbook*. Mill Valley, CA: University Science, 1989.
- [8] D. P. Kingma and M. Welling, “Auto-encoding variational Bayes,” 2013, arXiv:1312.6114. [Online]. Available: <https://arxiv.org/abs/1312.6114>

- [9] S. Liu, “Wi-Fi Energy Detection Testbed (12MTC),” 2023, gitHub repository. [Online]. Available: <https://github.com/liustone99/Wi-Fi-Energy-Detection-Testbed-12MTC>
- [10] “Treatment episode data set: discharges (TEDS-D): concatenated, 2006 to 2009.” U.S. Department of Health and Human Services, Substance Abuse and Mental Health Services Administration, Office of Applied Studies, August, 2013, DOI:10.3886/ICPSR30122.v2
- [11] K. Eves and J. Valasek, “Adaptive control for singularly perturbed systems examples,” Code Ocean, Aug. 2023. [Online]. Available: <https://codeocean.com/capsule/4989235/tree>